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Multiple origins of dorsal ecdysial sutures in trilobites and their relatives

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Abstract

Euarthropods are an extremely diverse phylum in the modern, and have been since their origination in the early Palaeozoic. They grow through moulting the exoskeleton (ecdysis) facilitated by breaking along lines of weakness (sutures). Artiopodans, a group that includes trilobites and their non-biomineralizing relatives, dominated arthropod diversity in benthic communities during the Palaeozoic. Most trilobites – a hyperdiverse group of tens of thousands of species - moult by breaking the exoskeleton along cephalic sutures, a strategy that has contributed to their high diversity during the Palaeozoic. However, the recent description of similar sutures in early diverging non-trilobite artiopodans mean that it is unclear whether these sutures evolved deep within Artiopoda, or convergently appeared multiple times within the group. Here we describe new well-preserved material of *Acanthomeridion*, a putative early diverging artiopodan, including hitherto unknown details of its ventral anatomy and appendages revealed through CT scanning, highlighting additional possible homologous features between the ventral plates of this taxon and trilobite free cheeks. We used three coding strategies treating ventral plates as homologous to trilobite free cheeks, to trilobite cephalic doublure, or independently derived. If ventral plates are considered homologous to free cheeks, *Acanthomeridion* is recovered sister to trilobites however dorsal ecdysial sutures are still recovered at many places within Artiopoda. If ventral plates are considered homologous to doublure or non-homologous, then *Acanthomeridion* is not recovered as sister to trilobites, and thus the ventral plates represent a distinct feature to trilobite doublure/free cheeks.

eLife assessment

The authors present 16 new well-preserved specimens from the early Cambrian Chengjiang biota. These specimens potentially represent a new taxon which could be **useful** in sorting out the problematic topology of artiopodan arthropods - a topic of interest to specialists in Cambrian arthropods. The authors provide **solid** anatomical and phylogenetic evidence in support of a new interpretation of the homology of dorsal sutures in trilobites and their relatives.

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1. Introduction

Euarthropods, more specifically members of the clade Artiopoda, dominated diversity in Palaeozoic animal communities (Daley et al., 2018). This group, which includes trilobites and their non-biomineralizing relatives, are united by a morphology consisting of a headshield covering a pair of uniramous antennae and at least three pairs of biramous limbs, followed by a dorsoventrally flattened body with a series of biramous appendages (Stein and Selden, 2012; Giribet and Edgecombe, 2019). The origins of key morphological features facilitating the diversity of trilobites can be found in their non-biomineralizing artiopodan relatives (e.g. Du et al., 2019; Schmidt et al., 2022). The cephalon, as an informative and functionally specialized body region, affords pivotal data for resolving the phylogenetic relationships and evolutionary history of arthropods, artiopodans, (Budd and Telford, 2009; Yang et al., 2013; Strausfeld et al., 2022) and trilobites in particular (e.g. Stubblefield 1936; Lieberman 2002; Paterson et al. 2019).

The presence, shape, and character of sutures in the trilobite cephalon have been central to classification schemes of the group for nearly 200 years (e.g. Emmrich 1839; Salter 1864; Beecher 1897a, b; Stubblefield 1936; Rasetti 1945; Whittington et al. 1997), while the similarity in the cranidial outlines of Cambrian suture-bearing groups supported a single origin or strong evolutionary constraint on its origins (Foote 1991; Hughes 2007). These sutures provided lines of weakness along which the cephalon split during moulting, creating an anterior ecdysial gape through which the animal exited the old exoskeleton (e.g. Henningsmoen 1975; Whittington 1990; Daley and Drage, 2016). Sutures may have served additional purposes (Stubblefield 1936) and were under additional functional demands, including feeding (Fortey & Owens 1999) and burrowing behaviours (Esteve et al. 2021). Different trilobite clades display distinct suture patterns, with two sets, the circumocular and marginal sutures, relevant for the creation of this ecdysial gape (Henningsmoen 1975). For the earliest diverging trilobites, olenellines (Palmer & Repina 1993; Lieberman 2002) which a have been recovered as a paraphyletic grade rather than a monophyletic group (e.g. Paterson et al. 2019) circumocular sutures facilitated shedding of the cornea during ecdysis (e.g. Ramskold & Edgecombe 1991) while a marginal suture (**Fig. 1e**) facilitated anterior egression through separation of the lateral cephalic doublure from the dorsal cephalon (e.g. Stubblefield 1936; Henningsmoen 1975). In later diverging trilobites the circumocular and marginal sutures combined into a single system (**Fig. 1d**). Here the cornea is fused to the free cheek (e.g. Stubblefield 1936; Ramsköld & Edgecombe 1991) and these facial sutures – the fused circumocular and marginal sutures - facilitated both ecdysis of the visual surface and anterior egression of the exoskeleton following withdrawal from the free cheeks (e.g. Henningsmoen 1975).

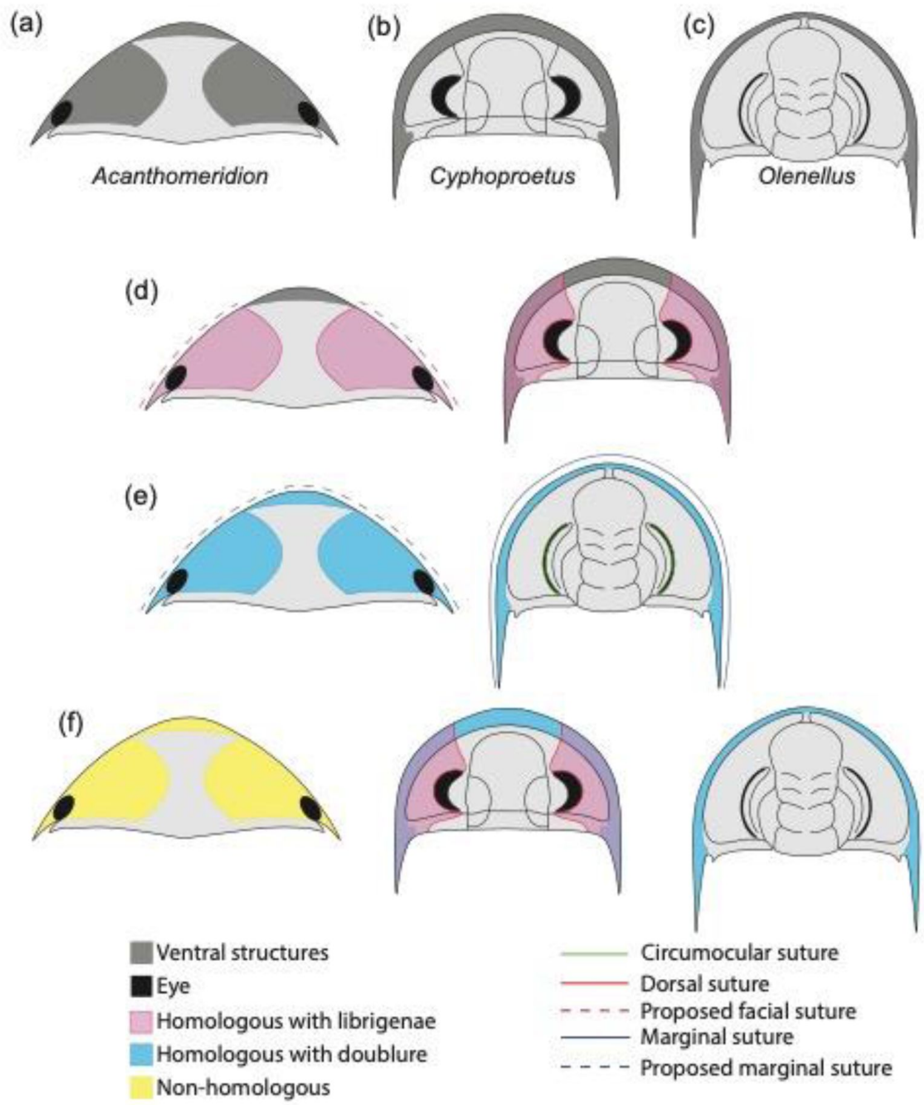

Fig. 1

Hypotheses of homology in sutures between *Acanthomeridion* and trilobites.

(a) Cephalic morphology of *Acanthomeridion serratum*. (b) Cephalic morphology of a proetid trilobite (modified from Daley and Drage, 2016). (c) Cephalic morphology of a redlichid trilobite (modified from Whittington 1989). (d) Ventral plates and librigenae as homologous with dorsal suture. (e) Ventral plates and doublure as homologous with marginal suture. (f) Non-homology between ventral plates of *Acanthomeridion*, librigenae and doublure.

The origin of these fused dorsal ecdysial sutures – facial sutures – has traditionally been considered to fall within Trilobita (e.g. Fortey and Whittington, 1989; Edgecombe and Ramsköld, 1999; Lieberman, 2002). However, the presence of dorsal ecdysial sutures in earlier diverging artiopodans – the Protosutura (Du et al. 2019) and eye slits in Petalopleura (Chen et al., 2019) raised questions of the homology and origins within Artiopoda (e.g. Hou et al. 2017; Du et al. 2019). An alternative hypothesis for the origins of the dorsal cephalic sutures emerged: that these had a deep root within Artiopoda and were subsequently lost in some groups including multiple times within Trilobita (Hou et al., 2017a; Du et al., 2019), rather than the traditional view that dorsal cephalic sutures in trilobites were derived within the clade, and thus these eye slits and facial sutures were acquired independently. Support for the deep root hypothesis comes from the variability of facial sutures in trilobites, and the recognition that convergence and loss of features is common when they have a function or allow adapation to a particular niche (e.g. Moore & Wilmer 1997). Further evidence for the variability of suture morphology comes from ontogenetic studies, such as the fusion of dalmanitinid facial sutures during ontogeny (Drage et al., 2018), and the loss of this feature in other trilobite such as '*Cedaria*' *woosteri* (Hughes et al. 1997). This indicates that there is scope for a facial suture to have been lost repeatedly in artiopodan groups earlier diverging than Trilobita, should this feature prove to have a deeper origin within Artiopoda (Hou et al. 2017; Du et al. 2019). A deep root for facial sutures within Artiopoda would have repercussions for the importance of the facial sutures in determining trilobite relationships (e.g. Jell 2003) and the position of a grade of olenelline trilobites as the earliest diverging members of the group (e.g. Paterson et al. 2019). To date, a third possibility – that the ventral plates of *Acanthomeridion* are homologous to the doublure of olenellids, and thus *Acanthomeridion* and olenelline trilobites share presence of an unfused marginal suture – has not received broad attention in the literature, but should also be considered in light of the similarities in position of the ventral plates and olenelline doublure, and the marginal rather than dorsal position of the suture in *Acanthomeridion*..

Resolving the complex history of dorsal ecdysial sutures evolution within Artiopoda, requires the nature of cephalic sutures in artiopodans such as *Acanthomeridion*, protosuturans, and petalopleurans to be resolved, and their phylogenetic position confidently determined (Hou et al., 2017a; Du et al., 2019). However, the morphology of many earlier diverging artiopodans is not completely understood (Ortega Hernández et al., 2013; Lerosey-Aubril et al., 2017), contributing a significant amounts of missing data to character matrices used in phylogenetic analyses. Up to now, *Acanthomeridion* was arguably the least well known of these early diverging artiopodans, with its ventral anatomy and non-biomineralized structures very poorly known (Hou and Bergström, 1997; Hou et al., 2017a).

Here we describe new specimens of *Acanthomeridion serratum*, using micro-CT to reveal details of the ventral anatomy, eyes, and appendages for the first time, and reconstruct the changing shape of the body during ontogeny. We use these new data to show that *A. anacanthus* is a junior synonym of *A. serratum*. We then assess the phylogenetic position of the species considering two hypotheses relating to the dorsal ecdysial sutures – as homologous to the dorsal ecdysial sutures (facial sutures combined of a fused marginal and circumocular sutures) of trilobites (**Fig. 1d**), that they are homologous to marginal sutures of olenellines (**Fig. 1e**), and that these structures are unique to *Acanthomeridion* (**Fig. 1f**), to evaluate the different hypothesis of origination within Artiopoda.

2. Materials and methods

New specimens of *Acanthomeridion* were collected from Jiucun town, Chengjiang (**Fig. 2**) and accessioned at the Management Committee of the Chengjiang Fossil Site World Heritage (CJHMD 00052– 00062), and Research Center of Paleobiology, Yuxi Normal University (YRCP 0016–0020). New specimens of *Wutingaspis tingi* (YRCP 0021) and *Xandarella spectaculum* (YRCP 0022) were collected from Haikou town, Kunming (**Fig. 2**) and housed at Research Center of Paleobiology, Yuxi Normal University. The holotype of *Zhiwenia coronata* (YKLP 12370) is deposited at the Key Laboratory for Palaeobiology, Yunnan University which was excavated from Xiaoshiba area, Kunming (**Fig. 2**). The specimens were photographed by a LEICA DFC 550 digital camera mounted to a Stereoscope LEICA M205 C, and scanned with a Zeiss Xradia 520 Versa X-ray Microscope and Nikon XTH 225 ST micro-focus X-ray tomography machine. All micro-CT images were processed with the software Drishti (Zhai et al., 2019). Figures were processed with CorelDRAW X8, Inkscape 1.0 and Illustrator.

Given that specimens with thorax longitudinally curved and specimens with flattened thorax are considered as two species of *Acanthomeridion*, despite having the same exoskeletal morphology, we measured their axial length and attempted to determined their true maximum width in all specimens, including those described in this study and previously documented measurable specimens. The data was initially analyzed using a bivariate regression fitting function in Microsoft Office Excel. Subsequently, the software Past (Hammer et al. 2001) was employed for testing and validation, obtaining the same function and conducting a rational analysis of the function.

The character matrix used for the phylogenetic analyses was adapted from Schmidt et al., (2022), itself adapted from Du et al. (2019) and Chen et al. (2020). *Olenellus getzi* was added, to facilitate comparison with olenelline trilobites, while *Anacheirurus adserai*, *Cryptolithus tesselatus* and *Olenoides serratus* were added to provide more trilobite terminals. *Olenellus getzi* was chosen as it represents the olenelline with best known soft tissues (antennae only; Dunbar 1925). Trunk appendages are known in *A. adserai, C. tesselatus* and *O. serratus* (Walcott, 1911; Whittington, 1975), with descriptions by Bicknell et al., 2021; Losso and Ortega-Hernández 2022 for coding the matrix. Our updated dataset includes 70 taxa (*Acanthomeridion anacanthus* was removed, four trilobites were added). Three matrices were used to test the alternative hypotheses of sutures, one with 100 characters considering the ventral plates unique, and not homologous to trilobite librigenae ('ventral plates under head' as a new character in the dataset) and two with 99 characters. The first of these considered the ventral plates homologous to trilobite librigenae, and the second to the doublure of olenellines (see Morphobank project). We added the character state 'ringed distribution lamellae' to Ch. 15 in all matrices (Chen et al., 2019; Du et al., 2019), and altered the characters relating to the suture pattern (Ch. 31, 32) to include character states suitable for *Olenellus getzi*. Given the difficulties of identifying circumocular sutures in non-biomineralizing artiopodans, Ch. 31 indicated the presence or absence of ecdysial sutures in the cephalon, while Ch. 32 indicated the type of suture. This could either be marginal and/or circumocular, but unfused (state 0) or marginal and circumocular fused into a facial suture (state 1). This facilitated differentiation between the state in *Olenellus getzi* and the other trilobites in the dataset, and thus allowed the two hypotheses relating to homology between the ventral plates and the doublure or free cheeks of trilobites to be tested. Eight additional characters relating to the trilobite glabellar region were added, taken from Paterson et al. (2019), in order to provide data to distinguish internal trilobite relationships. These were Ch. 5, 6, 7, 10, 11, 12, 14 and 15 in Paterson et al. (2019), which are Ch. 37 – 44 in all three matrices used for this study. A further character (calcified thorax separated into prothorax and opisthothorax) was also added (Ch. 99 or 100 in matrices used for this study). Matrices used in phylogenetic analyses are provided through morphobank (Project #P4290. Go to *morphobank.org*, then Log In. Use credentials: Email address: P4290, Password: Acanthomeridion2023).

Phylogenetic analyses were performed with MrBayes (Bayesian Inference) (Ronquist et al., 2012). Both the 'maximum information' and 'minimum assumptions' strategies of ref (Bapst et al., 2018) were utilised on an earlier version of the matrix, in order to confirm that model choice was not a major driver in differences for the results (Du et al. 2023). For the final version, the 'maximum information' strategy was used, as model choice did not play a large influence on the

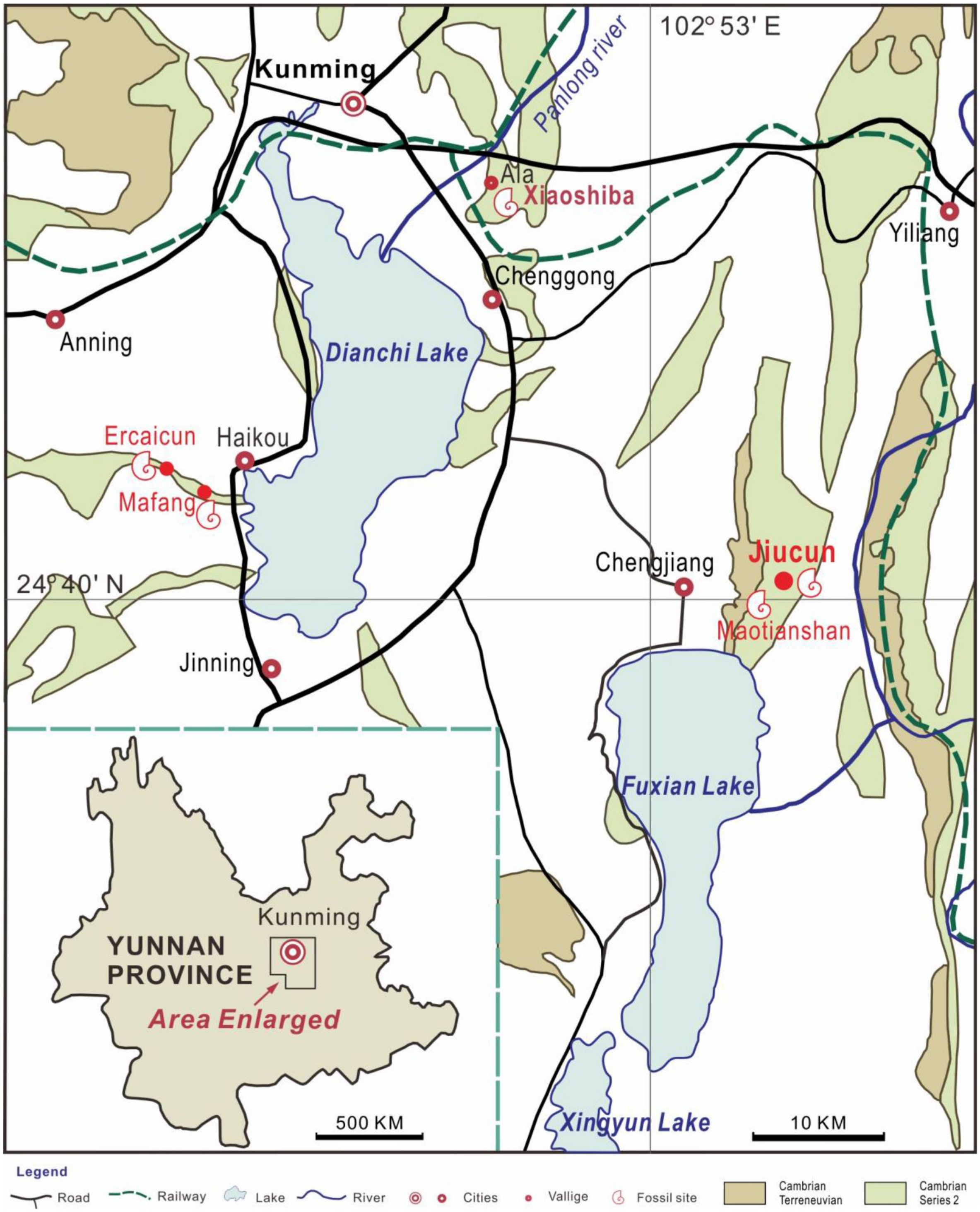

Fig. 2

Sites yielding *Acanthomeridion* from the Cambrian Stage 3 Chengjiang Biota (red circles).

All the specimens of *A. serratum* used here are collected from Jiucun town, Chengjiang County.

topologies recovered. Bayesian analyses were run for 20 million generations using four chains, every 1000th sample stored and 25% burn-in (Lewis, 2001; Ronquist et al., 2012). Convergence was diagnosed using Tracer (Rambaut et al., 2018).

For each coding strategy, two sets of Bayesian analyses were run. The first was unconstrained, whereas for the second all trilobites except for *Olenellus getzi* were constrained as a monophyletic group to recover the relationships of trilobites (*Olenellus* as the earliest diverging member) comparable to the comprehensive trilobite phylogenetic analysis of Paterson et al. (2019). Resulting posterior tree samples were compared by comparisons in the consensus trees, and in multidimensional treespace (using R code and method from Pates et al. 2022). The number of trees where *Acanthomeridion serratum* was sister to trilobites was quantified for each coding strategy and for constrained and unconstrained analyses, and the area of treespace occupied by the posterior samples – and trees where *A. serratum* was sister to triobites within those samples – were visualized.

3. Results

3.1 Systematic palaeontology

ARTIOPODA Hou and Bergström, 1997

Acanthomeridion Hou et al., 1989

Type and only species

Acanthomeridion serratum Hou et al., 1989

Diagnosis

Non-biomineralized artiopodan with elongate dorsal exoskeleton that gives body an elliptical outline. Subtriangular head shield with deep lateral notches that accommodate stalked elliptical eyes, rounded genal angles of the head shield, an axe-like hypostome, and paired teardrop-shaped plates on either side of hypostome. Head bears large eyes and long multi-segmented antennae consisting of over 40 podomeres anterior to three pairs of small cephalic limbs. Trunk composed of 11 tergites bearing expanded tergopleurae with well-developed distal spines, and a terminal spine. The ninth tergite bears a pleural spine more elongate than others, eleventh tergite reduced and expanded into leaf-like outline. Each tergite bears a pair of biramous limbs with long and dense spines on the endopods, and slender stick-like exopods with long and dense bristles. Modified from Hou et al., 1989, 2017a.

Acanthomeridion serratum Hou et al., 1989 **Syn. nov.**

Figs. 3–5; **7g-i**; S1–6

1989 *Acanthomeridion serratum* Hou et al., pl. Ⅲ, figs. 1-5; pl. Ⅳ, figs. 1-5; p. 46, text-figs. 3, 4.

1996 *Acanthomeridion serratum* Chen et al., p. 157, figs 199-203.

1997 *Acanthomeridion serratum* Hou and Bergström, p. 38.

1999 *Acanthomeridion serratum* Luo et al., p. 51, pl. 6 fig. 5.

1999 *Acanthomeridion serratum* Hou et al., p. 130, fig. 188.

2004 *Acanthomeridion serratum* Chen, p. 280, fig. 442.

2004 *Acanthomeridion serratum* Hou et al., p. 176, fig. 16.63; p. 177, fig. 16.64.

2017a *Acanthomeridion serratum*, *A. anacanthus*, and *Acanthomeridion* sp in Hou et al., p. 734, figs. 1-4.

2017b *Acanthomeridion serratum* Hou et al., p. 204, fig. 20.34; p. 205, fig. 20.35.

Type material

Holotype, CN 108305. *Paratype*, CN 108306–108310.

New material

Sixteen new specimens were collected from Jiucun (**Fig. 2**), Chengjiang County and housed in the Management Committee of the Chengjiang Fossil Site World Heritage (CJHMD 00052–00062), and Research Center of Paleobiology, Yuxi Normal University (YRCP 0016–0020).

Occurrence

Yu'anshan Member, Qiongzhusi Formation, *Wutingaspis–Eoredlichia* biozone, Cambrian Stage 3. Jiucun and Maotianshan, Chengjiang County, Yunnan, China; Mafang and Ercaicun, Haikou town, Kunming, Yunnan, China (**Fig. 2**).

Diagnosis

As for genus, by monotypy.

Description

The dorsal exoskeleton of *Acanthomeridion serratum* displays weak trilobation and elliptical outline. Specimens measure from 20 to 75 mm along the sagittal axis (**Figs. 3a, c**; **4a**; **5**; S1–6). The head shield is subtriangular in dorsal view, with rounded anterior margin and small spines on posterior margin (**Figs. 3a, c**; **4a**; **5**; S6a, c). A pair of lateral posterior notches accommodate stalked elliptical eyes. The lateral margin of the head shield is a suture that attaches paired ventral plates. Ventral plates have a teardrop outline, occupy the entire length of the cephalon and terminate in a posterior spine (maximum length 3.4 mm; Fig. S2b) that projects into the thorax as far as the anteriormost tergite (Fig. S1a). The medial margin is curved towards the axially line and sits adjacent to a conterminant axe-shaped hypostome (maximum dimensions 9.6 mm x 11.6 mm; **Fig. 3a, b**). Four pairs of appendages are present in the head (**Figs. 3f–j**; S5a). A pair of antennae with at least 43 articles (**Figs. 4a, b**; S2a, f; S4b) sits anterior to three pairs of small cephalic limbs (**Fig. 3f–j**). The cephalic appendages are poorly preserved and only the gross morphology can be discerned as no individual podomeres are visible. The protopodite of the cephalic limbs are subtriangular in outline. Endopodites and exopodites are not clearly preserved. Appendage three may include the endopodite, but no podomeres are visible.

The trunk is composed of 11 tergites which extend laterally into spinose tergopleurae. Tergites are all subequal in length. Tergites 8-11 curve towards the posterior, and reduce in width progressively, so that T11 is approximately 25% the width of T8 and 33% the width of T9 (**Figs. 3a, c**; **4a**; S1a, c, d). Each tergite has small spines on its lateral and posterior margins (**Figs. 6**; S1a, c–f; S2i, k; S3a, j; S5b, e, f; S6a–f). The shape of tergites and the convexity of the trunk changes through ontogeny (**Fig. 5**). In the smaller specimens, the trunk appears slender from a dorsal view as the body displays a high convexity (**Fig. 5t**). Pleurae curve ventrally, and only slightly towards the posterior (**Fig. 5a–k**). In larger specimens, the trunk is less convex than for the

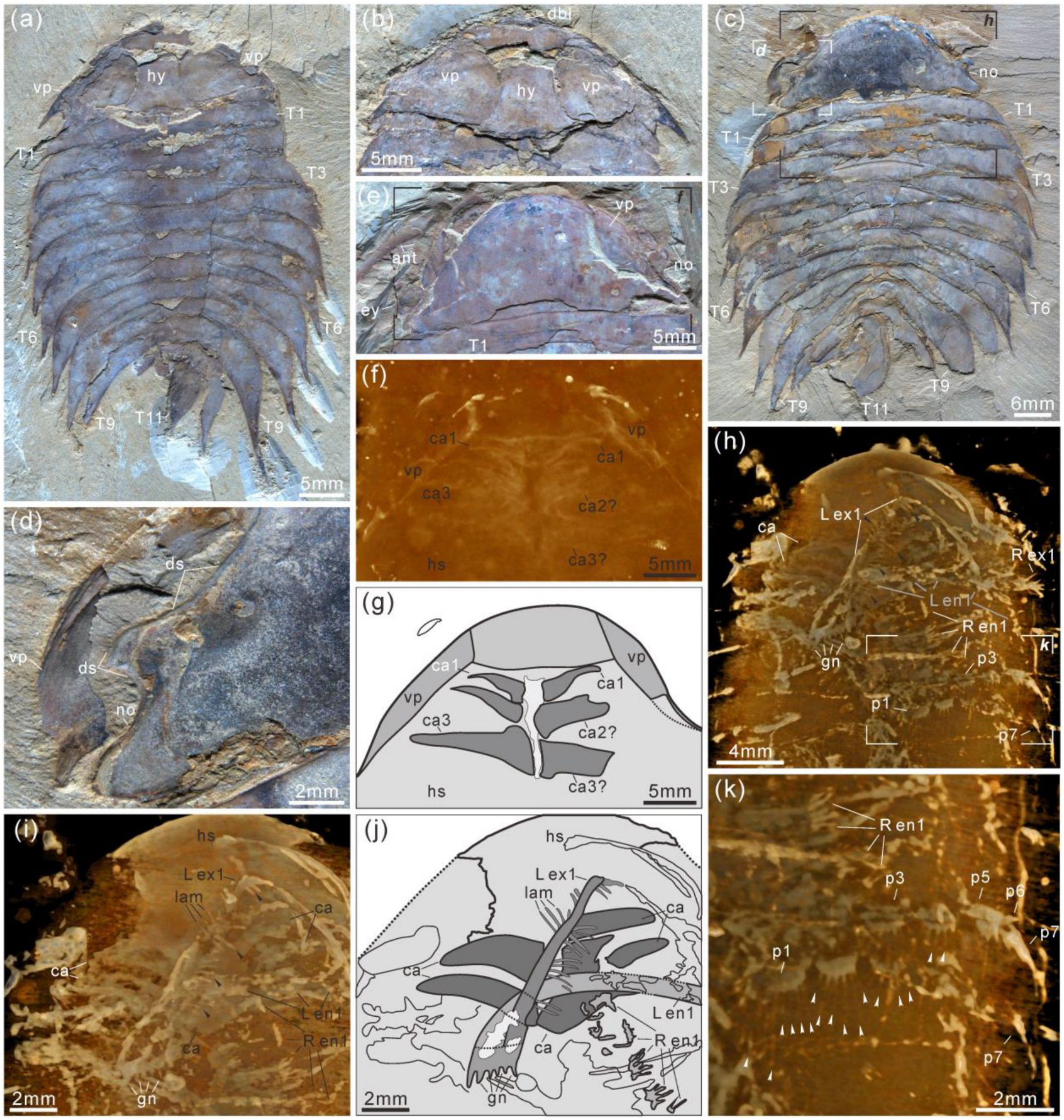

Fig. 3

Acanthomeridion serratum from the Cambrian Stage 3 Chengjiang Biota.

(a, b) CJHMD 00052a/b respectively, individual with hypostome, ventral plates, and 11 tergites. (c, d) CJHMD 00053a, showing ventral plate, dorsal sutures, and 11 tergites. (e–g) YRCP 0016a, showing the ventral plates, and three post-antennal limbs. (h–k) Showing the post-antennal appendages under head, gnathobases of trunk limbs, stick-like exopodites with bristles (black arrows), and endopodites with long spines (white arrows). (f) Micro-CT image of YRCP 0016a; (h, i, k) Micro-CT images of CJHMD 00053a. Abbreviations: ant, antenna; ca*n*, post-antennal appendage *n* beneath head; dbl, doublure; ds, dorsal suture; en, endopodites; ex, exopodites; ey, eye; hs, head shield; hy, hypostome; L, left; lam, lamellae; no, notch; p*n*, podomere *n*; R, right; T*n*, tergite *n*; ts, terminal spine; vp, ventral plate.

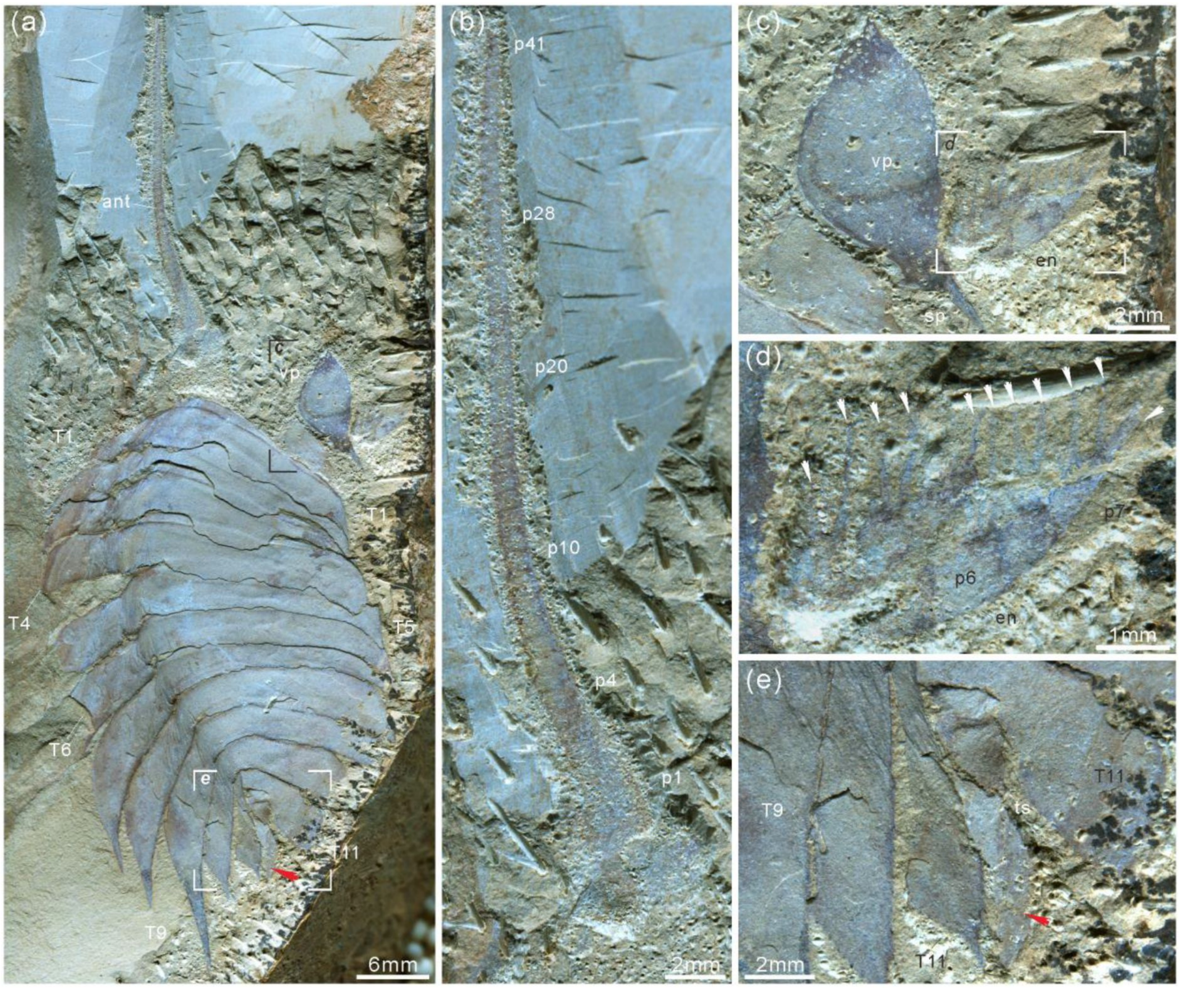

Fig. 4

***Acanthomeridion serratum* from the Cambrian Stage 3 Chengjiang Biota.**

(a–e) CJHMD 00055, showing the antenna, ventral plate, endopodites with long spines (arrows in d), 11 tergites, and paddle-like structure (red arrow). (a) Overview of whole specimen. (b) Detail of long left antenna. (c) Close-up of ventral plate. (d) Details of long spines (arrows) of right endopodites. (e) Close-up of paddle-like structure (red arrow). Abbreviations same as **Figs. 2**, 3.

smaller specimens (**Fig. 5s**), ventral curvature of pleurae is decreased, and posterior curvature increased (**Fig. 5o–r**). A linear relationship between trunk axial length and width is obtained (**Fig. 6**). Where trunk appendages are preserved, each tergite is associated with a pair of biramous trunk appendages (**Figs. 3a, c, h, k**; **4**; S1a, b). The protopodite is sub triangular with a broad attachment to the body wall and short robust gnathobasal spines along the medial margin (**Figs. 3h–j**; S3a–f, i). The ventral margin of the protopodite has endites longer than the gnathobasal spins. The dorsal edge of the protopodite is poorly preserved and does not clearly show the attachment of the exopodite. Exopodites are slender, stick-like and long (8.3 mm, ca. 3x protopodite length), bearing two rows of lamellae (**Figs. 3h, k**; S1a, b). Exopods are shorter than the width of corresponding tergites (**Figs. 3h, S1a, b**). The lamellae appear short but may not be completely preserved. There is no evidence that the exopodite divided into lobes or has articulations. Endopodites are composed of seven trapezoidal podomeres. Podomeres 1–6 bear spiniferous endites arranged in rows, six on pd1, 2, four on pd3, 4, and three on pd5, 6. Podomere seven displays a terminal claw of three elongate spines (**Figs. 3h, k**, k; **4d**). From the preserved appendages, no significant antero-posterior differentiation or dramatic size reduction can be discerned (**Figs. 3**, S1) One specimen shows evidence for paired midgut diverticulae (Fig. S4a, f). The body terminates in a long, slender spine, dorsal to a paired paddle-like structure (**Figs. 4a, e**; S5b, g).

Remarks

The eyes were previously identified as genal projections (see **Fig. 1**, **Fig. 2A**, and **Fig. 3A** in Hou et al., 2017a) because of the limited available specimens and the posterior location of the eyes. *Acanthomeridion anacanthus* Hou et al. 2017a is here proposed to be a junior subjective synonym of *Acanthomeridion serratum* Hou et al.1989, an interpretation supported by:

1. The paddle-like structure under the terminal spine (**Figs. 4a, e**; S5b, g) was previously interpreted as the twelfth tergite. The specimen YKLP 11115 (see **Fig.4** in Hou et al., 2017a) which is a similar size to YKLP 11116 (holotype of *Acanthomeridion anacanthus*) and YKLP 11113 (see **Fig.2A**, C and **Fig.3A**, C in Hou et al., 2017a) show 11 tergites.
2. Specimens CJHMD 00052-55 and YRCP 0016 show the evidently pleural spines of T8 and T9 (**Figs. 3a, c**; **4a**; S1a, c, d; S2a, f; S4b, g).
3. The orientation of pleurae makes some specimens appear narrower than others. Specifically, those where pleurae curve ventrally look more slender than those where pleurae are curved posteriorly. Specimen YRCP 0017 preserves right ventrally curving pleura and left flat pleura, which lead to its right pleura narrower than the left (**Figs. 5j, o**; S2i). Moreover, most small specimens (1cm–3cm) are preserved with ventrally curving pleurae (**Figs. 5a–e**; S3a–c; S4a; S5a-c; S6g, i) and these appear narrower than the few small specimens with flat pleurae (**Figs. 5f**; S4b. See also Fig.2A–D, and Fig.4 in Hou et al., 2017a). Larger specimens are typically preserved with flattened, posteriorly curving pleurae.
4. Twenty specimens show a good ontogenetic series of *Acanthomeridion* including some that have previously have been assigned to '*Acanthomeridion anacanthus*' (**Figs. 5**, **6**). Linear measurements do not support distinct species of different sizes, nor with different proportions of thorax axial length to width. The length and width data fit well with a linear equation; most of the data fall within the 95% forecast (**Fig. 6b**). Data fall close to the fitted line, as indicated by the R^2 value of 0.95. Lastly, specimens previously assigned to '*A. anacanthus*' are recovered as smaller *A. serratum* specimens with similar thoracic shape.

The new specimens and CT data presented herein allow the description of cephalic and trunk limbs, and a conterminant hypostome adjacent to ventral plates in *Acanthomeridion* for the first time. The presence of a terminal spine on these ventral plates is identified. The ventral plate is separated from the dorsal part of the cephalon by a suture that runs close to the stalked eyes. The possible affinities of these ventral plates, and the phylogenetic implications of these, are discussed

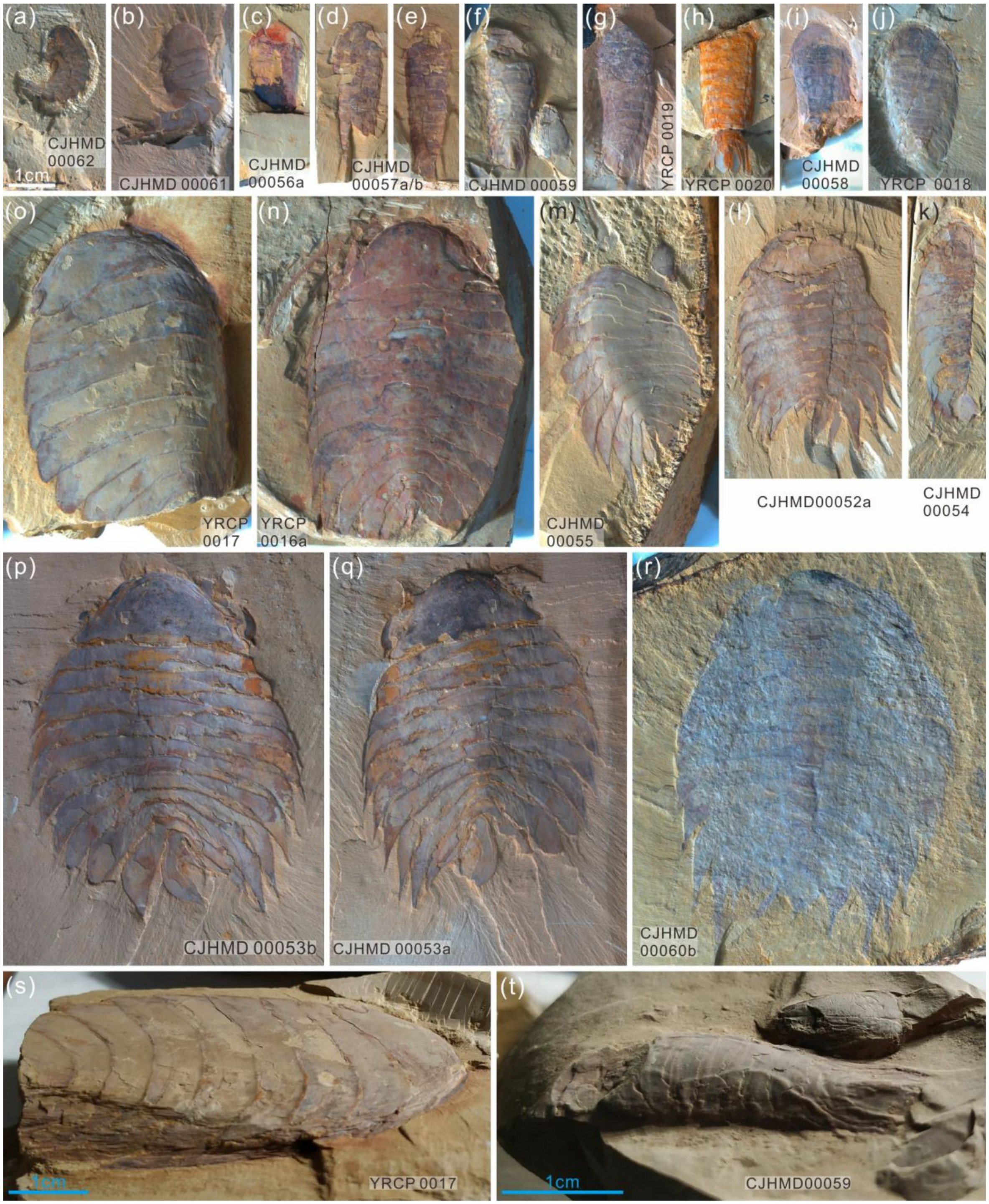

Fig. 5

Ontogenetic series (a–r) of *Acanthomeridion* and their ventrally curling pleurae (s, t).

(a–r) Showing the individuals from smallest to largest with same scale bar. (s) Lateral view of (o), note the right curling pleurae and left flat pleurae. (t) Lateral view of (f), showing the left curling pleurae.

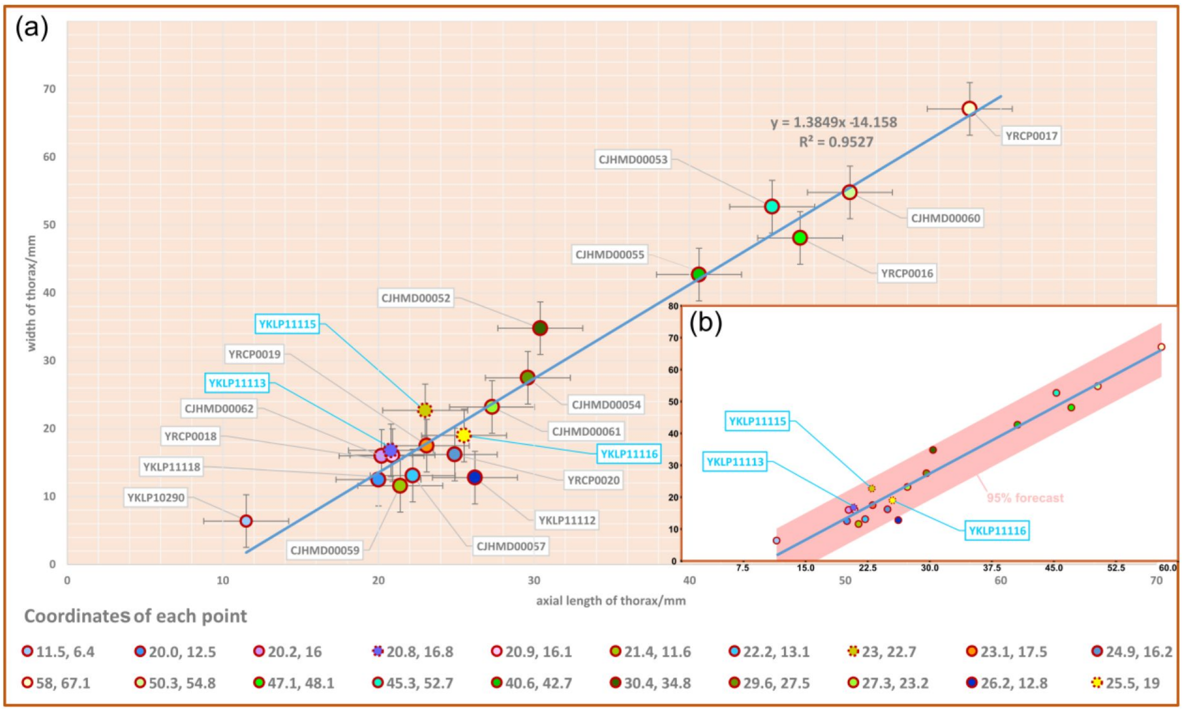

Fig. 6

Scatterplot of axial length and width of thorax of *Acanthomeridion serratum*.

(a) scatterplot of raw data. (b) data with 95% forecast shown in pink. Specimens previously assigned to *A. anacanthus* indicated by blue lettering.

below (Subsections 3.2, 3.3) The presence of four appendages in the head (or three in the head and one underneath the articulation) is comparable to *Zhiwenia* (Du et al., 2019), as well as other artiopodans including trilobites (Edgecombe and Ramsköld, 1999). However, the morphology of the three non-antenniform cephalic limbs is difficult to determine with only the protopodite clearly visible. The lack of observed endopodites and exopodites on these three appendages may be a preservational artefact. Possibly the absence of clear endopodites and exopodites represents specialization of the cephalic appendages with the structures being significantly reduced. Other Cambrian artiopodans have been shown to have appendage specialization (Losso and Ortega-Hernández, 2022) which is frequently found in the cephalon (Chen et al., 2019; Schmidt et al., 2022), but none show reduction of both the exopodite or endopodite. In *Sinoburius lunaris*, the first two post-antennal appendages have reduced endopodites with elongated exopodites (Chen et al., 2019), and in *Pygmaclypeatus daziensis* the exopodite is significantly reduced in the cephalic appendages (Schmidt et al., 2022).

Recently, the head of *Retifacies abnormalis* was interpreted to develop three pairs of uniramous post-antennal appendages and following by one biramous appendage pair (Zhang et al., 2022). It would be unclear the function of an appendage with both distal elements being significantly reduced until more clear appendages are found.

The exopodites of *Acanthomeridion* also differ substantially from those known in other artiopodans. Specifically, the stick-like nature of the exopodite, the rows of lamellae combined with the apparent lack of segmentation, differ from the paddle shaped exopodites thought to characterise the remainder of the group (e.g. Schmidt et al., 2022, and **fig. 4** in Ortega Hernández et al., 2013).

3.2 Possible affinities of the ventral plates in *Acanthomeridion*

3.2.1. Ventral plates as homologues of trilobite free cheeks

The ventral plates of *Acanthomeridion* have previously been suggested to be homologues to the free cheeks of trilobites, with the suture between these features and the remainder of the cephalon interpreted as a facial suture (**Fig. 1d**) (e.g. Hou et al. 2017). Morphological similarities are apparent between the ventral plates and free cheeks in many trilobites (**Fig. 7a, b**) (Whittington et al., 1997) as well as the path of suture separating them from the rest of the cephalon. Our new data provide additional support for interpreting the ventral plates as homologues of the librigenae. The ventral plates of *Acanthomeridion* are here shown to be transversely broad, tapering posteriorly to a spine, in a position directly comparable to the genal spine of many trilobites (**Figs. 1d**; **7a, b**). Furthermore, the ventral plate is separated from the rest of the cephalon by a suture that passes near the compound eye. This path is similar to the facial suture (fused circumocular and marginal sutures) of many trilobites especially those bearing opisthoparian facial sutures, which separate the free cheeks from the cranidium passing through the visual surface (**Fig. 7a, b**) (Whittington et al., 1997). The homology of eye notches (e.g., *Luohuilinella* and *Zhiwenia* (**Fig. 7e, f**)), and eye slits (e.g. *Phytophilaspis* (Ivantsov, 1999) of petalopleurans, dorsal ecdysial sutures in trilobites and *Acanthomeridion* has been suggested previously (Du et al. 2019). These eye slits join the dorsal eye with the anterolateral margin of the head shield, possibly forming a single suture that facilitated release of the visual surface and created an anterior gape for the animal to leave the old exoskeleton during ecdysis. However, the evolution from eye notches to eye slits and finally to the loss of the eye slit can been seen, calling into question the affinity between these structures and sutures (Chen et al. 2019). This hypothesis was tested explicitly in the first matrix, where the ventral plates of *Acanthomeridion* were considered homologous to the free cheeks of trilobites. In this analysis, the dorsal slits and sutures of other artiopodans were also treated as homologous ecdysial sutures in the cephalon (**Fig. 8a**).

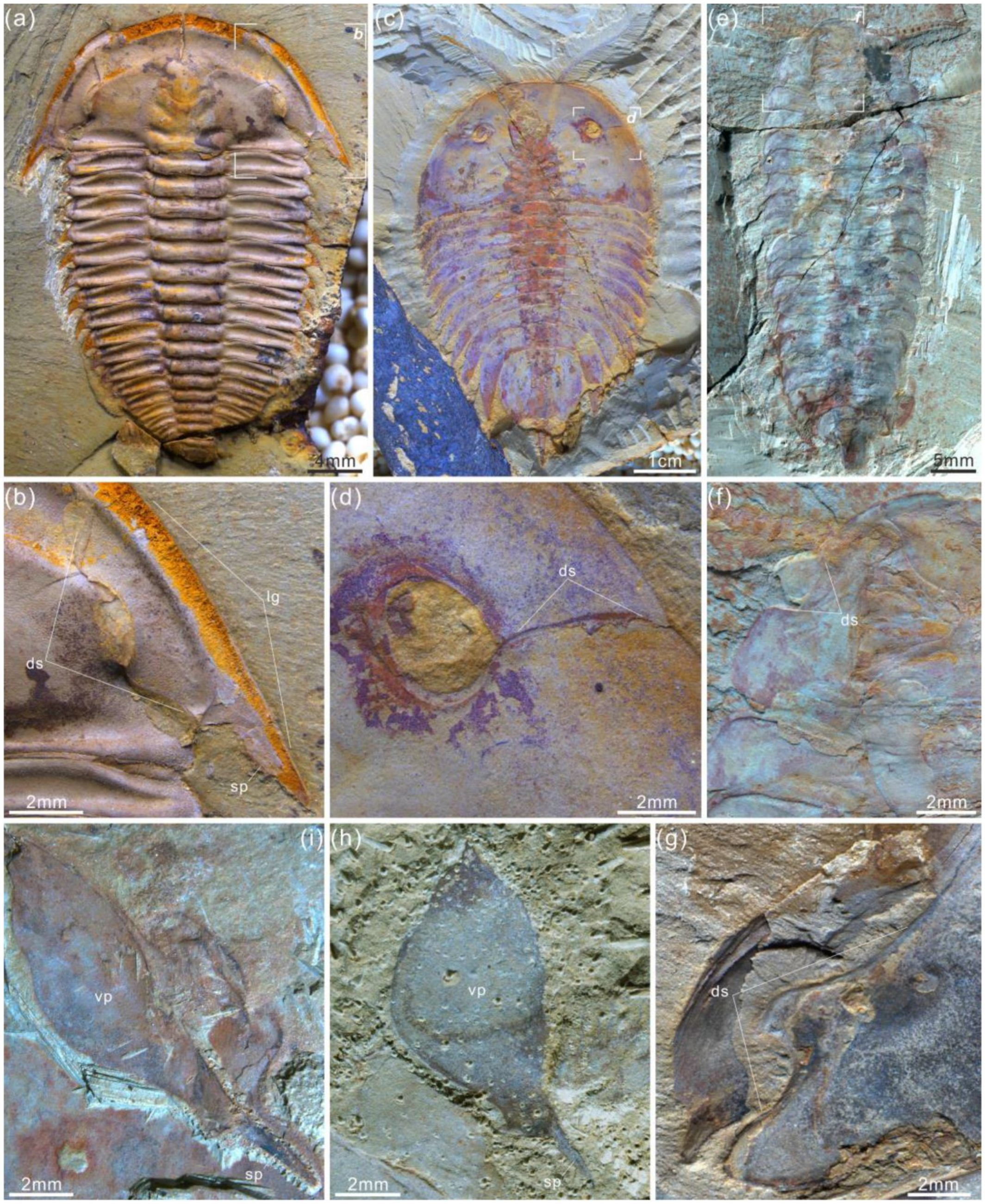

Fig. 7.

Four representative artiopodans from early Cambrian with their dorsal sutures, free cheeks and ventral plates.

(a, b) The iconic trilobite *Wutingaspis tingi* from Chengjiang Biota, note its free cheeks and dorsal or facial sutures. (c, d) The symbolic xandarellid artiopodan *Xandarella spectaculum* from Chengjiang Biota bearing the distinctive dorsal sutures. (e, f) Holotype of the protosuturan artiopodan *Zhiwenia coronata* from Xiaoshiba Biota developing dorsal sutures. (g) The left dorsal suture of *Acanthomeridion serratum* from Chengjiang Biota, showing the morphological and positional similarities to that of *W. tingi* (a, b), *X. spectaculum* (c, d), and *Z. coronata* (e, f). (h, i) Right ventral plates of *A. serratum* from Chengjiang Biota bearing a terminal spine, which is similar to free cheek of trilobite like *W. tingi* (a, b).

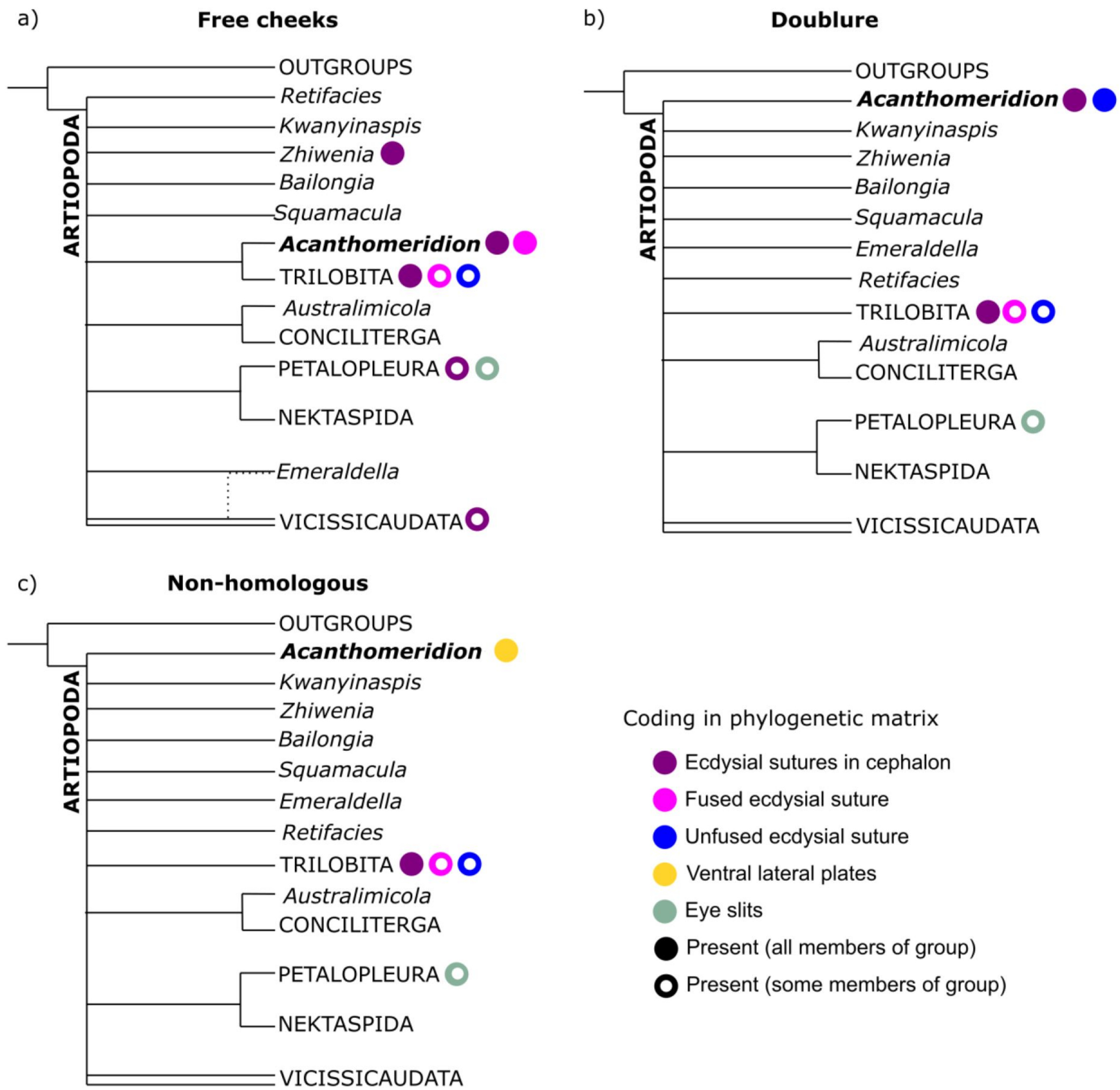

Fig. 8.

Simplified results of phylogenetic analyses.

Comparison of results from matrices with different coding strategies, with coding of key characters for terminals within the analysis indicated by colored circles. (a) Ventral plates considered homologous to free cheeks. (b) Ventral plates considered homologous to trilobite doublure. (c) Ventral plates not considered homologous to any artiopodan character.

3.2.2. Ventral plates as homologues to doublure in olenelloid trilobites

In the above scenario, the ventral plates of *Acanthomeridion* would be homologous to the free cheeks both dorsally and ventrally in trilobites, including the doublure. However it is possible that the ventral plates are only homologous to the doublure, and thus that the suture represents the marginal suture, rather than a fused facial suture. The difficulty lies in the marginal position of the eyes, which could support the interpretation that these are fused sutures, but might also represent a position that facilitated ecdysis through a single, unfused, suture. Considered homologous only to the doublure, the suture in *Acanthomeridion* separating the ventral plates from the cephalon would be homologous to the unfused, marginal suture in olenelloids rather than the fused facial suture of most other trilobites (**Fig. 1e**) (e.g. Stubblefield 1936).

Few morphological features can be drawn to support this hypothesis beyond the ventral position of the plates. In particular, the lateral position of the eyes in *Acanthomeridion*, and the path of the suture close to these eyes, suggests that the suture functioned to release both the ventral plate and the eye. Although some artiopodans are known with expansive doublure such as *Squamacula buckorum* (Paterson et al., 2012) this structure usually follows the outline of the margin and lacks spines (Whittington et al., 1997).

This hypothesis was tested in the second matrix, where the ecdysial suture of *Acanthomeridion* was considered unfused (**Fig. 8b**). In this scenario, the presence of eye slits and other dorsal sutures in other artiopodans would not represent homologous structures, as the proposed evolution is from a ventral-lateral structure to a dorsal one. Thus, the eye slits of the petalopleurans who bear them were not considered as ecdysial sutures for this analysis.

3.2.3 Ventral plates and eye slits of artiopodans as unique structures without homologues in trilobites

Eye slits of petalopleuran artiopodans as homologues to facial sutures of trilobites has drawn support from comparisons such as *Loganopeltoides* (Rasetti 1948), where a suture charts a path from the dorsal eye to the anterolateral margin of the head shield. However, in *Loganopeltoides* this suture represents the vestiges of the free cheeks (Edgecombe & Ramksold 1999), a derived state within Trilobita. Given that the features in support of homology between the ventral plates of *Acanthomeridion* and trilobite free cheeks, and implicit within this a deep root of the facial suture within Artiopoda rather than Trilobita, may not represent true homologues (e.g. spines are often evolved convergently, and the dorsal eye slits of artiopodans have previously been considered unique characters – see e.g. Edgecombe & Ramskold 1999; Chen et al., 2019) it is important to consider the hypothesis that these features are unique to *Acanthomeridion*. Thus for this third analysis, the ventral plates of *Acanthomeridion* were treated as their own character in the matrix (**Fig. 8c**).

3.3 Phylogenetic analyses

All six Bayesian analyses produced the same broad relationships within Artiopoda, with the exception of the position of *Acanthomeridion* (**Fig. 8**). Petalopleurans were recovered sister to Nektaspida, *Australimicola* sister to Conciliterga, and Vicissicaudata was not completely resolved. The positions of *Bailongia, Emeraldella, Kwanyinaspis, Squamacula* and *Zhiwenia* were not resolved, and the internal relationships of Artiopodans were also not well resolved. *Acanthomeridion* was recovered as sister to Trilobita when the ventral plates were considered homologous to the free cheeks of trilobites, but otherwise its position was not well resolved. Within Trilobita, when the analysis was unconstrained, *Olenellus* and *Eoredlichia* formed a group sister to the other trilobites, or these two taxa were recovered in a polytomy with a clade containing the other trilobites (SFigs. 8—10). When the group *Anacherirurus, Cryptolithus, Eoredlichia, Olenoides* and *Triarthrus* was constrained (with the exclusion of *Olenellus*), then

Olenellus was recovered as sister to all other trilobites (SFigs. 8—10). This result is important for considering the hypothesis that the ventral plates of *Acanthomeridion* were homologous to the doublure of trilobites.

When visualised in treespace, constrained and unconstrained analyses produce very similar results (**Fig. 9**). Trees in the posterior sample of the analysis where the ventral plates of *Acanthomeridion* were treated as homologous to the free cheeks of trilobites fall in the area of treespace supporting *Acanthomeridion* as sister to trilobites. This result is corroborated by the raw counts in the full sample (**Table 1**), and by the support for this node as seen in the consensus tree (**Fig. 9**). Posterior samples of analyses treating ventral plates as homologous to the doublure and those treating them as non-homologous with any trilobite feature overlapped in treespace, over a much broader region. These display a very similar confidence ellipse to the sample of trees where *Acanthomeridion* is not resolved as sister to trilobites, but instead in a different position in the tree (**Fig. 9**). Again, these treespace results are corroborated by the raw counts in the full sample (**Table 1**) and by the poorly resolved position of *Acanthomeridion* in the consensus trees (**Fig. 9**).

4. Discussion and conclusion

All consensus trees, of parsimony and Bayesian results from both coding strategies, support multiple origins of cephalic sutures in Artiopoda (**Fig. 8**, SFigs. 8—10). When the sutures of *Acanthomeridion* and trilobites are considered as homologous (e.g., Hou et al., 2017), as well as to those of other artiopodans (e.g., Du et al., 2019), terminals with ecdysial sutures in the cephalon are found in multiple groups in Artiopoda (**Fig. 8a**), although *Acanthomeridion* is recovered sister to trilobites. If instead the sutures in *Acanthomeridion* are considered marginal, and homologous to those of olenelline trilobites, *Acanthomeridion* and trilobites do not form a clade (**Fig 8b**, **Table 1**), and trees occupy a different region of the treespace (**Fig. 9**). Indeed, the posterior samples of analyses treating the ventral plates as not homologous to any other artiopodan feature (**Fig 8c**) are extremely similar to those treating them as homologous to the doublure of trilobites, as illustrated by their broad overlap in treespace (**Fig. 9**) and similar support for nodes in the consensus trees (SFigs. 8—10). Thus, two of the three coding strategies receive some support from the phylogenetic results. Either *Acanthomeridion* shares the presence of facial sutures with trilobites, and these form a clade, but the sutures in petalopleurans and *Zhiwenia* are distinct (multiple origins of dorsal ecdysial sutures) or the ventral lateral plates of *Acanthomeridion* are a distinct feature to the free cheeks of trilobites (and petalopeluran and *Zhiwenia* features are also distinct; multiple origins of sutures). The position of *Acanthomeridion* as not sister to trilobites in the analyses treating the ventral plates as homologous to doublure mean that this hypothesis did not gain support, given the assumptions and character matrices of this study.

Support for *Acanthomeridion* as sister to trilobites comes from morphological similarity between the spinose termination of the ventral plates of *Acanthomeridion* (similar to trilobite genal spines) and the suture that follows an oblique line and passes near the eye both for *Acanthomeridion* (similar to trilobite opisthoparian facial sutures; Whittington et al., 1997). Taken together with trilobate exoskeleton of large *Acanthomeridion* specimens, the non-biomineralised *Acanthomeridion* bears many morphological similarities to biomineralised trilobites that further support a sister relationship. However, this sister relationship rests on the treatment of the ventral plates as free cheeks. If this interpretation is not favored, then these similarities can be considered convergent, and the position of *Acanthomeridion* within Artiopoda is not well resolved by these analyses. For this interpretation of the ventral plates, the path of the suture between the ventral plates and cephalon passing close to the eye may represent a way to minimise damage to the eye during ecdysis, which is comparable to what is observed in trilobites with facial sutures.

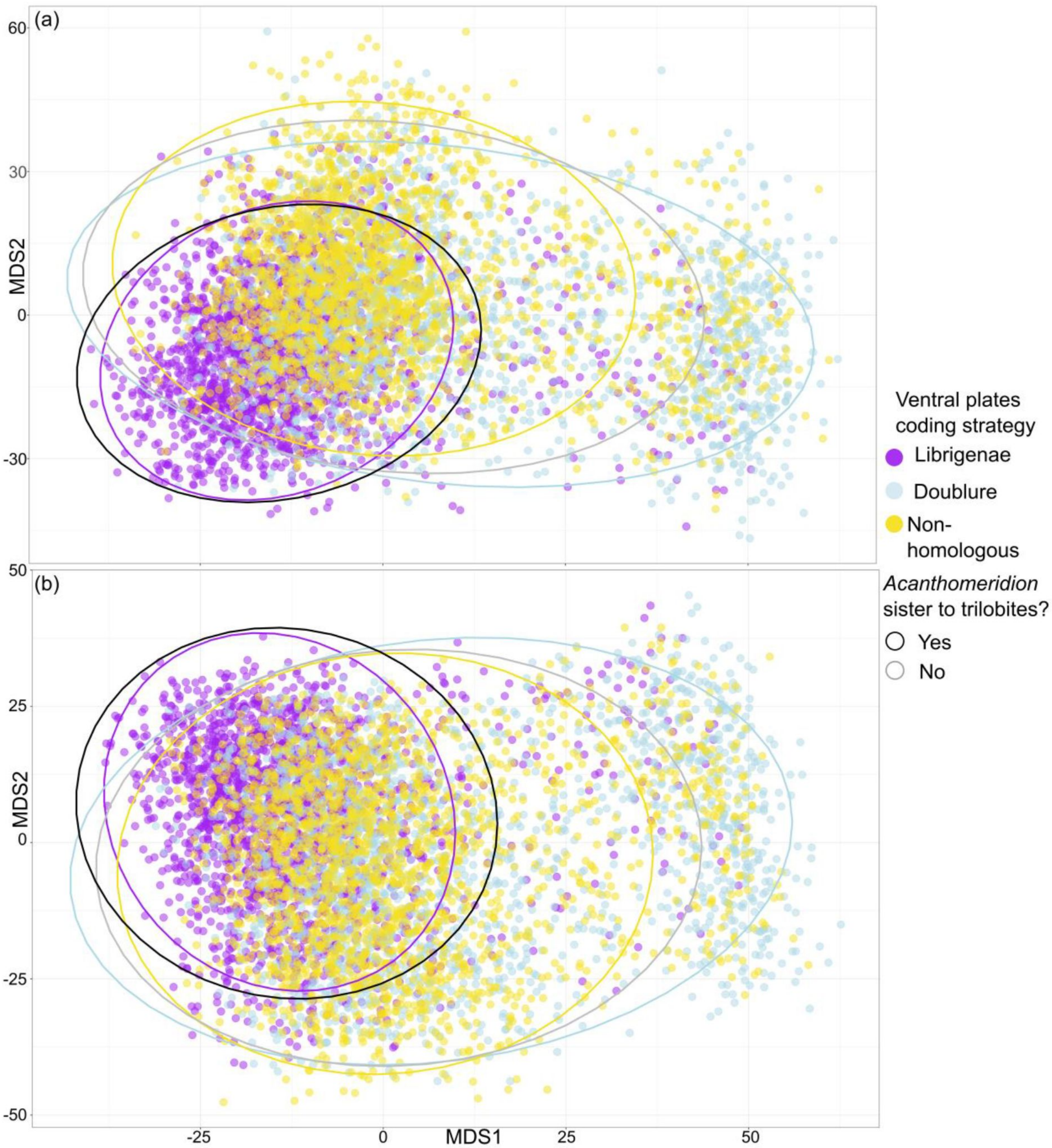

Figure 9.

Treespace visualisation of Bayesian Inference phylogenetic analyses.

Multidimensional treespace plotted by bipartition relating to coding strategies (color of point). (a) Analyses were unconstrained. (b) Trilobites were constrained so that *Olenellus getzi* was resolved as the earliest diverging member of the group. Area included within a 95% confidence interval (CI) indicated by points inside the line of different colors. Area where *Acanthomeridion* is recovered sister to trilobites (95% CI) also shown by area within black line. Where *Acanthomeridion* is not recovered sister to trilobites (95% CI), shown by area within grey line. Note the overlap in the areas of the black and purple lines.

	Unconstrained	Constrained
Homologous to free cheeks	7832 / 10504 (75%)	8083/10504
Homologous to doublure	1496 / 12004 (12%)	1836/10504
Not homologous	761 / 12004 (6%)	926/10504

Table 1

number of trees in posterior sample supporting *Acanthomeridion* as sister to trilobites / number of trees in the posterior sample total.

Results tabulated for each coding strategy, with unconstrained and constrained analyses (constrained analyses where a monophyletic group of all trilobites except for *Olenellus getzi* was forced, in order to recover trilobite relationships compatible with Paterson et al. 2019).

Other features observed in *Acanthomeridion* for the first time – uniramous deutocerebral appendages, an axe-shaped conterminant hypostome, two librigenal-like plates and three additional head appendages – are consistent with a phylogenetic position for the species as an early diverging artiopodan, or as sister to trilobites. Multipodomerous uniramous deutocerebral antennae are known in nearly all other members of the clade (Zhang et al., 2007; Du et al., 2019; Zhai et al., 2019), and thus this feature is not very informative. Similarly, a conterminant hypostome was already known in numerous other artiopodans (Zhang et al., 2004; Stein and Selden, 2012). As this type of hypostomal attachment is considered ancestral within Trilobita (Fortey, 1990), its presence in the possible sister to trilobites (if dorsal sutures are considered homologous) is consistent with these expectations from trilobite anatomy. If *Acanthomeridion* is instead an early diverging artiopodan, the presence of natant hypostomes in both trilobitomorphs and vicissicaudates (Fortey, 1990; Chen et al., 2019) demonstrates that a natant hypostome has most likely evolved repeatedly within Artiopoda.

In summary, new fossils and CT-scan data reveals the ventral anatomy and appendages of *Acanthomeridion serratum* for the first time, and additional specimens demonstrate that *A. 'anacanthus'* represents a junior synonym of *A. serratum,* with some features thought to distinguish the two instead ontogenetic in origin. The presence of a posteriorly orientated spine on the cephalic ventral plates of *Acanthomeridion* might be homologous to the genal spine of trilobite librigenae, providing additional support for a.sister group relationship between these taxa. Phylogenetic analyses interrogating this hypothesis recover a sister relationship between *Acanthomeridion* and trilobites, but still require multiple origins of cephalic sutures within the group. If the ventral plates are instead treated as homologous to the doublure of trilobites, or as not homologous to any trilobite feature, then no close relationship is recovered, and the morphological similarities between *Acanthomeridion* and trilobites such as the path of the suture separating the ventral plates from the cephalon, and details of the ventral plates, should instead be treated as convergent. Thus, regardless of how the ventral plates of *Acanthomeridion* are interpreted, multiple origins of cephalic sutures in artiopodans are recovered by phylogenetic analyses.

Conflict of interest

The authors declare that they have no conflict of interest.

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CRediT authorship contribution statement

Kunsheng Du: Conceptualization, Data Curation, Investigation, Methodology, Visualization, Writing – original draft, Writing – review and editing. **Jin Guo:** Resources, Data curation, Validation. **Sarah R. Losso:** Conceptualization, Data Curation, Formal analysis, Writing – original draft, Writing – review and editing. **Stephen Pates:** Conceptualization, Data Curation, Investigation, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review and editing, Supervision. **Ming Li:** Methodology, Data curation. **Ailin Chen:** Resources, Funding acquisition, Project administration, Visualization, Supervision.

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Editors

Reviewing Editor

Ariel Chipman

The Hebrew University of Jerusalem, Jerusalem, Israel

Senior Editor

George Perry

Pennsylvania State University, University Park, United States of America

Reviewer #1 (Public Review):

Summary:

Du et al. report 16 new well-preserved specimens of atiopodan arthropods from the Chengjiang biota, which demonstrate both dosal and vental anatomies of a pothential new taxon of atiopodans that are closely related to trolobites. Authors assigned their specimens to Acanthomeridion serratum, and proposed A. anacanthus as a junior subjective synonym of Acanthomeridion serratum. Critially, the presence of ventral plates (interpreted as cephalic liberigenae), together with phylogenic results, lead authors to conclude that the cephalic sutures originated multiple times within the Artiopoda.

Strengths:

New specimens are highly qualified and informative. The morphology of dorsal exoskeleton, except for the supposed free cheek, were well illustrated and described in detail, which provide a wealth of information for taxonmic and phylogenic analyses.

Weaknesses:

The weaknesses of this work is obvious in a number of aspects. Technically, ventral morphlogy is less well revealed and is poorly illustrated. Additional diagrams are necessary to show the trunk appendages and suture lines. Taxonomically, I am not convinced by authors' placement. The specimens are markedly different from either Acanthomeridion serratum Hou et al. 1989 or A. anacanthus Hou et al. 2017. The ontogenetic description is extremely weak and the morpholical continuity is not established. Geometric and morphomitric analyses might be helpful to resolve the taxonomic and ontogenic uncertainties. I am confused by author's description of free cheek (libragena) and ventral plate. Are they the same object? How do they connect with other parts of cephalic shield, e.g. hypostome and fixgena. Critically, homology of cephalic slits (eye slits, eye notch, doral suture, facial suture) not extensivlely discussed either morphologically or functionally. Finally, authors claimed that phylogenic results support two separate origins rather than a deep origin. However, the results in Figure 4 can be explain a deep homology of cephalic suture in molecular level and multiple co-options within the Atiopoda.

Comments on the revised version:

I have seen the extensive revision of the manuscript. The main point "Multiple origins of dorsal ecdysial sutures in atiopoans" is now partially supported by results presented by the authors. I am still unsatisfied with descriptions and interpretations of critical features newly revealed by authors. The following points might be useful for the author to make further revisions.

(1) The antennae were well illustrated in a couple of specimens, while it was described in a short sentence.
(2) There are also imprecise descriptions of features.
(3) Ontogeny of the cephalon was not described.
(3) The critical head element is the so called "ventral plate". How this element connects with

the cephalic shield is not adequately revealed. The authors claimed that the suture is along the cephalic margin. However, the lateral margin of cephalon is not rounded but exhibit two notches (e.g. Fig 3C) . This gives an indication that the supposed ventral plates have a dorsal extension to fit the notches. Alternatively, the "ventral plate" can be interpreted as a small free cheek with a large ventral extension, providing evidence for librigenal hypothesis.

https://doi.org/10.7554/eLife.93113.2.sa2

Reviewer #3 (Public Review):

Summary:

Well-illustrated new material is documented for Acanthomeridion, a formerly incompletely known Cambrian arthropod. The formerly known facial sutures are proposed be associated with ventral plates that the authors homologise with the free cheeks of trilobites (although also testing alternative homologies). An update of a published phylogenetic dataset permits reconsideration of whether dorsal ecdysial sutures have a single or multiple origins in trilobites and their relatives.

Strengths:

Documentation of an ontogenetic series makes a sound case that the proposed diagnostic characters of a second species of Acanthomeridion are variation within a single species. New microtomographic data shed light on appendage morphology that was not formerly known. The new data on ventral plates and their association with the ecdysial sutures are valuable in underpinning homologies with trilobites.

I think the revision does a satisfactory job of reconciling the data and analyses with the conclusions drawn from them. Referee 1's valid concerns about whether a synonymy of Acanthomeridion anacanthus is justified have been addressed by the addition of a length/width scatterplot in Figure 6. Referee 2's doubts about homology between the librigenae of trilobites and ventral plates of Acanthomeridion have been taken on board by re-running the phylogenetic analyses with a coding for possible homology between the ventral plates and the doublure of olenelloid trilobites. The authors sensibly added more trilobite terminals to the matrix (including Olenellus) and did analyses with and without constraints for olenelloids being a grade at the base of Trilobita. My concerns about counting how many times dorsal sutures evolved on a consensus tree have been addressed (the authors now play it safe and say "multiple" rather than attempting to count them on a bushy topology). The treespace visualisation (Figure 9) is a really good addition to the revised paper.

Weaknesses:

The question of how many times dorsal ecdysial sutures evolved in Artiopoda was addressed by Hou et al (2017), who first documented the facial sutures of Acanthomeridion and optimised them onto a phylogeny to infer multiple origins, as well as in a paper led by the lead author in Cladistics in 2019. Du et al. (2019) presented a phylogeny based on an earlier version of the current dataset wherein they discussed how many times sutures evolved or were lost based on their presence in Zhiwenia/Protosutura, Acanthomeridion and Trilobita. The answer here is slightly different (because some topologies unite Acanthomeridion and trilobites). This paper is not a game-changer because these questions have been asked several times over the past seven years, but there are solid, worthy advances made here.

I'd like to see some of the most significant figures from the Supplementary Information included in the main paper so they will be maximally accessed. The "stick-like" exopods are not best illustrated in the main paper; their best imagery is in Figure S1. Why not move that figure (or at least its non-redundant panels) as well as the reconstruction (Figure S7) to the main paper? The latter summarises the authors' interpretation that a large axe-shaped hypostome appears to be contiguous with ventral plates. The specimens depict evidence for three pairs of post-antennal cephalic appendages but it's a bit hard to picture how they functioned if there's no room between the hypostome and ventral plates. Also, a comment is required on the reconstruction involving all cephalic appendages originating against/under the hypostome rather the first pair being paroral near the posterior end of the hypostome and the rest being post-hypostomal as in trilobites.

https://doi.org/10.7554/eLife.93113.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***eLife assessment***
>
> *The authors present 16 new well-preserved specimens from the early Cambrian Chengjiang biota. These specimens potentially represent a new taxon which could be useful in sorting out the problematic topology of artiopodan arthropods - a topic of interest to specialists in Cambrian arthropods. Because the anatomic features in the new specimens were neither properly revealed nor correctly interpreted, the evidence for several conclusions is inadequate.*

We thank the Senior Editor, Reviewing Editor and three reviewers for their work, and for their comments aimed at improving this project and manuscript. We have engaged with all the comments in detail, in order to strengthen our work. This includes adding additional data to support that all *Acanthomeridion* specimens belong to a single species, running further phylogenetic analyses including more trilobite terminals to test the specific hypothesis and interpretation raised by Reviewer 2, and visualising our results in treespace in order to determine support for the different interpretations of the ventral structures and their implications for the evolution of Artiopoda. We have also greatly expanded the introduction, which we feel adds clarity to areas misunderstood by some reviewers in the previous version of the manuscript.

Our point-by-point response to the public reviews of the reviewers are outlined below. We have also made changes resulting from the additional suggestions which are not public, which we have not reproduced below. We submit a new version of the main text, and can provide a tracked changes version if required. The new main text includes 9 figures and is 8624 words including captions and reference list.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *Du et al. report 16 new well-preserved specimens of atiopodan arthropods from the Chengjiang biota, which demonstrate both dorsal and ventral anatomies of a potential new taxon of artipodeans that are closely related to trilobites. Authors assigned their specimens to* Acanthomeridion serratum *and proposed* A. anacanthus *as a junior subjective synonym of* Acanthomeridion serratum*. Critically, the presence of ventral plates (interpreted as cephalic liberigenae), together with phylogenic results, lead authors to conclude that the cephalic sutures originated multiple times within the Artiopoda.*

We thank Reviewer 1 for their comments on the strengths and weaknesses of the previous version of the manuscript. We hope that the revised version strengthens our conclusions that *Acanthomeridion anacanthus* is a junior synonym of *A. serratum.*

> *Strengths:*
>
> *New specimens are highly qualified and informative. The morphology of the dorsal exoskeleton, except for the supposed free cheek, was well illustrated and described in detail, which provides a wealth of information for taxonomic and phylogenic analyses.*
>
> *Weaknesses:*
>
> *The weaknesses of this work are obvious in a number of aspects. Technically, ventral morphology is less well revealed and is poorly illustrated. Additional diagrams are necessary to show the trunk appendages and suture lines. Taxonomically, I am not convinced by the authors' placement. The specimens are markedly different from either Acanthomeridion serratum Hou et al. 1989 or A. anacanthus Hou et al. 2017. The ontogenetic description is extremely weak and the morpholical continuity is not established. Geometric and morphometric analyses might be helpful to resolve the taxonomic and ontogenic uncertainties.*

We appreciate that the reviewer was not convinced by our synonimisation in the first version of the manuscript. The recommendation of the reviewer to provide linear morphometric support for our synonymisation was much appreciated. We have provided measurements of the length and width of the thorax (Figure 6 in the new version), visualising the position of specimens previously assigned to *A. anacanthus*, to show this morphological continuity. These act as a complement to Figure 5, which shows the fossils in an ontogenetic trend.

> *I am confused by the author's description of the free cheek (libragena) and ventral plate. Are they the same object? How do they connect with other parts of the cephalic shield, e.g. hypostome, and fixgena? Critically, the homology of cephalic slits (eye slits, eye notch, dorsal suture, facial suture) is not extensively discussed either morphologically or functionally.*

We appreciate that the brevity of the introduction in the previous version led to some misunderstandings and some confusion. We have provided a greatly expanded introduction, including a new Figure 1, which outlines the possible homologies of the ventral plates and the three hypotheses considered in this study. The function of the cephalic and dorsal suture are now discussed in more detail both in introduction and discussion.

> *Finally, the authors claimed that phylogenic results support two separate origins rather than a deep origin. However, the results in Figure 4 can explain a deep homology of the cephalic suture at molecular level and multiple co-options within the Atiopoda.*

A deep molecular origin is difficult to demonstrate using solely fossil material from an extinct group such as Artiopoda. Thus our study focuses on morphological origins. The number of losses required for a deep morphological origin means that we favour multiple independent morphological origins.

> ***Reviewer #2 (Public Review):***
>
> *Overall: This paper describes new material of* Acanthomeridion serratum *that the authors claim supports its synonymy with* Acanthomeridion anacanthus*. The material is important and the description is acceptable after some modification. In addition, the paper offers thoughts and some exploration of the possibility of multiple origins of the*

> *dorsal facial suture among artiopods, at least once within Trilobita and also among other non-trilobite artiopods. Although this possibility is real and apparently correct, the suggestions presented in this paper are both surprising and, in my opinion, unlikely to be true because the potential homologies proposed with regard to* Acanthomeridion *and trilobite-free cheeks are unconventional and poorly supported.*
>
> *What to do? I can see two possibilities. One, which I recommend, is to concentrate on improving the descriptive part of the paper and omit discussion and phylogenetic analysis of dorsal facial suture distribution, leaving that for more comprehensive consideration elsewhere. The other is to seek to improve both simultaneously. That may be possible but will require extensive effort.*

We thank the reviewer for their detailed comments and suggestions for multiple ways in which we might revise the manuscript. We have taken the option that is more effort, but we hope more reward, in interrogating the larger question alongside improving the descriptive part of the paper. This has taken a long time and incorporation of new techniques, but has in our opinion greatly strengthened the work.

> *Major concerns*
>
> *Concern 1 - Ventral sclerites as free cheek homolog, marginal sutures, and the trilobite doublure*
>
> *Firstly, a couple of observations that bear on the arguments presented - the eyes of A. serratum are almost marginal and it is not clear whether a) there is a circumocular suture in this animal and b) if there was, whether it merged with the marginal suture. These observations are important because this animal is not one in which an impressive dorsal facial suture has been demonstrated - with eyes that near marginal it simply cannot do so. Accordingly, the key argument of this paper is not quite what one would expect. That expectation would be that a non-trilobite artiopod, such as A. serratum, shows a clear dorsal facial suture. But that is not the case, at least with A. serratum, because of its marginal eyes. Rather, the argument made is that the ventral doublure of A. serratum is the homolog of the dorsal free cheeks of trilobites. This opens up a series of issues.*

We appreciate that the reviewer disagrees with both interpretations we offered for the ventral plates, and has offered a third interpretation for the homology of this feature with the doublure of trilobites. Support for our original interpretation comes from the position of the eye stalks in *Acanthomeridion*, which fall very close to the suture between ventral plate rest of the cephalon. However, we appreciate that the reviewer has a valid interpretation, that the ventral plates might be homologues of the doublure alone.

To clarify the (two, now three) hypotheses of homology for the ventral plates considered in this study, we provide a new summary figure (Figure 1). In addition, the introduction has been greatly lengthened with further discussion of the different suture types in trilobites, their importance for trilobite classification schemes, and extensive references to older literature are now included. Further, we add background to the hypotheses around the origins of dorsal ecdysial sutures.

We add that the interpretation of *A. serratum* as having features homologous to the dorsal sutures of trilobites is already present in the literature, and so while the reviewer may disagree with it, it is certainly a hypothesis that requires testing.

> *The paper's chief claim in this regard is that the "teardrop" shaped ventral, lateral cephalic plates in Acanthomeridion serratum are potential homologs of the "free cheeks" of those trilobites with a dorsal facial suture. There is no mention of the possibility that*

> *these ventral plates in A. serratum could be homologs of the lateral cephalic doublure of olenelloid trilobites, which is bound by an operative marginal suture or, in those trilobites with a dorsal facial suture, that it is a homolog of only the doublure portions of the free cheeks and not with their dorsal components.*

We include this third possibility in our revised analyses and manuscript. To test this properly required adding in an olenelloid trilobite to our matrix, as we needed a terminal that had both a marginal and circumoral suture, but not fused. We chose *Olenellus getzi* for this purpose, as it is the only *Olenellus* with some appendages known (the antennae). We also added further characters to the morphological matrix, and additional trilobites from which soft tissues are known, in order to better resolve this part of the tree. Trilobites in the final analyses were: *Anacheirurus adserai, Cryptolithus tesselatus, Eoredlichia intermedia, Olenoides serratus, Olenellus getzi, Triarthrus eatoni.*

However, addition of these trilobites added a further complication. Under unconstrained analysis, *Olenellus getzi* was resolved with *Eoredlichia intermediata* as a clade sister to all other trilobites.

Thus the topology of Paterson et al. 2019 (*PNAS*) was not recovered, and so the hypothesis of Reviewer 2 could not be robustly tested. In order to achieve a topology comparable to Paterson et al., we ran a further three analyses, where we constrained a clade of all trilobites except for *O. getzi*. This recovered a topology where the earliest diverging trilobites had unfused sutures, and thus one suitable for considering the role of *Acanthomeridion serratum* ventral plates as homologues of the doublure of trilobites.

Unfortunately, for these analyses (both constrained and unconstrained), *Acanthomeridion* was not resolved as sister to trilobites, but instead elsewhere in the tree (see Table 1 in main text, Fig. 9, and SFig 9). Thus our analyses do not find support for the reviewer's hypothesis as multiple origins of this feature are still required.

It was still an excellent point that we should consider this hypothesis, and we have retained it, and discussion surrounding it, in our manuscript.

> *The introduction to the paper does not inform the reader that all olenelloids had a marginal suture - a circumcephalic suture that was operative in their molting and that this is quite different from the situation in, say, "Cedaria" woosteri in which the only operative cephalic exoskeletal suture was circumocular. The conservative position would be that the olenelloid marginal suture is the homolog of the marginal suture in A. serratum: the ventral plates thus being homolog of the trilobite cephalic doublure, not only potential homolog to the entire or dorsal only part of the free cheeks of trilobites with a dorsal facial suture. As the authors of this paper decline to discuss the doublure of trilobites (there is a sole mention of the word in the MS, in a figure caption) and do not mention the olenelloid marginal suture, they give the reader no opportunity to assess support for this alternative.*
>
> *At times the paper reads as if the authors are suggesting that olenelloids, which had a marginal cephalic suture broadly akin to that in Limulus, actually lacked a suture that permitted anterior egression during molting. The authors are right to stress the origin of the dorsal cephalic suture in more derived trilobites as a character seemingly of taxonomic significance but lines such as 56 and 67 may be taken by the non-specialist to imply that olenelloids lacked a forward egressionpermiting suture. There is a notable difference between not knowing whether sutures existed (a condition apparently quite common among soft-bodied artiopods) and the well-known marginal suture of olenelloids, but as the MS currently reads most readers will not understand this because it remains unexplained in the MS.*

As noted in response to a previous point (above) we now have a greatly expanded introduction which should give the reader an opportunity to assess support for this alternative hypothesis. We now include *Olenellus getzi* in our analyses, and have added characters to the morphological matrix to make this clear.

A reference to the case of '*Cedaria' woosteri* is made in the introduction to highlight further the variability of trilobites, as is a reference to Foote's analysis of cranidial shapes and support this provides for a single origin of the dorsal suture.

> *With that in mind, it is also worth further stressing that the primary function of the dorsal sutures in those which have them is essentially similar to the olenelloid/limulid marginal suture mentioned above. It is notable that the course of this suture migrated dorsally up from the margin onto the dorsal shield and merged with the circumocular suture, but this innovation does not seem to have had an impact on its primary function - to permit molting by forward egression. Other trilobites completely surrendered the ability to molt by forward egression, and there are even examples of this occurring ontogenetically within species, suggesting a significant intraspecific shift in suture functionality and molting pattern. The authors mention some of this when questioning the unique origin of the dorsal facial suture of trilobites, although I don't understand their argument: why should the history of subsequent evolutionary modification of a character bear on whether its origin was unique in the group?*

We include reference to evolutionary modification and loss of this character as it is important to stress that if a character is known to have been lost multiple times it is possible that it had a deeper root (in an earlier diverging member of Artiopoda than Trilobita) and was lost in olenelloids. This is the question that we seek to address in our manuscript.

> *The bottom line here is that for the ventral plates of A. serratum to be strict homologs of only the dorsal portion of the dorsal free cheeks, there would be no homolog of the trilobite doublure in A. serratum. The conventional view, in contrast, would be that the ventral plates are a homolog of the ventral doublure in all trilobites and ventral plates in artiopods. I do not think that this paper provides a convincing basis for preferring their interpretation, nor do I feel that it does an adequate job of explaining issues that are central to the subject.*

We stress that our interpretations – that the ventral plates are not homologous to any artiopodan feature or that they are homologous to the free cheeks of trilobites – have both been raised in the literature before. Whereas we could not find mention of the reviewer's 'conventional view' relating to *Acanthomeridion*. We appreciate that this view is still valid and worth investigating, which we have done in the further analyses conducted. However, we did not find support for it. Instead we find some support for both ventral plates as homologues of free cheeks, and as unique structures within Artiopoda.

> *Concern 2. Varieties of dorsal sutures and the coexistence of dorsal and marginal sutures*
>
> *The authors do not clarify or discuss connections between the circumocular sutures (a form of dorsal suture that separates the visual surface from the rest of the dorsal shield) and the marginal suture that facilitates forward egression upon molting. Both structures can exist independently in the same animal - in olenelloids for example. Olenelloids had both a suture that facilitated forward egression in molting (their marginal suture) and a dorsal suture (their circumocular suture). The condition in trilobites with a dorsal facial suture is that these two independent sutures merged - the formerly marginal suture migrating up the dorsal pleural surface to become confluent with the circumocular suture. (There are also interesting examples of the expansion of the circumocular suture*

> *across the pleural fixigena.) The form of the dorsal facial suture has long figured in attempts at higher-level trilobite taxonomy, with a number of character states that commonly relate to the proximity of the eye to the margin of the cephalic shield. The form of the dorsal facial suture that they illustrate in Xanderella, which is barely a strip crossing the dorsal pleural surface linking marginal and circumocular suture, is comparable to that in the trilobites Loganopeltoides and Entomapsis but that is a rare condition in that clade as a whole. The paper would benefit from a clear discussion of these issues at the beginning - the dorsal facial suture that they are referring to is a merged circumcephalic suture and circumocular suture - it is not simply the presence of a molt-related suture on the dorsal side of the cephalon.*

We have added in an expanded introduction where these points are covered in detail. We appreciate that this was not clear in the earlier version, and this suggestion has greatly improved our work.

> *Concern 3. Phylogenetics*
>
> *While I appreciate that the phylogenetic database is a little modified from those of other recent authors, still I was surprised not to find a character matrix in the supplementary information (unless it was included in some way I overlooked), which I would consider a basic requirement of any paper presenting phylogenetic trees - after all, there's no a space limit. It is not possible for a reviewer to understand the details of their arguments without seeing the character states and the matrix of state assignments.*

A link to a morphobank project was included in the first submission. This project has been updated for the current submission, including an additional matrix to treat the reviewer's hypothesis for the ventral plates. Morphobank Project #P4290. Email address: P4290, reviewer password:

Acanthomeridion2023, accessible at morphobank.org. We have added in additional details for the reviewer and others to help them access the project:

The project can be accessed at morphobank.org, using the below credentials to log in: Email address: P4290, Password: Acanthomeridion 2023.

> *The section "phylogenetic analyses" provides a description of how tree topology changes depending on whether sutures are considered homologous or not using the now standard application of both parsimony and maximum likelihood approaches but, considering that the broader implications of this paper rest of the phylogenetic interpretation, I also found the absence of detailed discussion of the meaning and implications of these trees to be surprising, because I anticipated that this was the main reason for conducting these analysis. The trees are presented and briefly described but not considered in detail. I am troubled by "Circles indicate presence of cephalic ecdysial sutures" because it seems that in "independent origin of sutures" trilobites are considered to have two origins (brown color dot) of cephalic ecdysial sutures - this may be further evidence that the team does not appreciate that olenelloids have cephalic ecdysial sutures, as the basal condition in all trilobites. Perhaps I'm misunderstanding their views, but from what's presented it's not possible to know that. Similarly, in the "sutures homologous" analyses why would there be two independent green dots for both Acanthomeridion and Trilobita, rather than at the base of the clade containing them both, as cephalic ecdysial sutures are basal to both of them? Here again, we appear to see evidence that the team considers dorsal facial sutures and cephalic ecdysial sutures to be synonymous - which is incorrect.*

We appreciate that the reviewer misunderstood the meaning of the dots, leading to confusion. The dots indicated how features were coded in the phylogenetic analysis. In our revised version of this figure (Figure 8 in the new version), these dots are now clearly labelled as indicating 'coding in phylogenetic matrix'. Further, with the revised character list, we now can provide additional detail for the types of sutures (relevant as we now include more trilobite terminals).

> *This point aside, and at a minimum, that team needs to do a more thorough job of characterizing and considering the variety of conditions of dorsal sutures among artiopods, their relationships to the marginal suture and to the circumocular suture, the number, and form of their branches, etc.*

We thank the reviewer for this summary, and appreciate their concerns and thorough review. Our revised version takes into account all these points raised, and they have greatly improved the clarity, scope and thoroughness of the work.

> ***Reviewer #3 (Public Review):***
>
> *Summary:*
>
> *Well-illustrated new material is documented for Acanthomeridion, a formerly incompletely known Cambrian arthropod. The formerly known facial sutures are shown to be associated with ventral plates that the authors very reasonably homologise with the free cheeks of trilobites. A slight update of a phylogenetic dataset developed by Du et al, then refined slightly by Chen et al, then by Schmidt et al, and again here, permits another attempt to optimise the number of origins of dorsal ecdysial sutures in trilobites and their relatives.*
>
> *Strengths:*
>
> *Documentation of an ontogenetic series makes a sound case that the proposed diagnostic characters of a second species of Acanthomeridion are variations within a single species. New microtomographic data shed some light on appendage morphology that was not formerly known. The new data on ventral plates and their association with the ecdysial sutures are valuable in underpinning homologies with trilobites.*

We thank the Reviewer 3 for their positive comments about the manuscript. We appreciate the constructive comments for improvements, and detailed corrections, which we have incorporated into our revised work.

> *Weaknesses:*
>
> *The main conclusion remains clouded in ambiguity because of a poorly resolved Bayesian consensus and is consistent with work led by the lead author in 2019 (thus compromising the novelty of the findings). The Bayesian trees being majority rules consensus trees, optimising characters onto them (Figure 7b, d) is problematic. Optimising on a consensus tree can produce spurious optimisations that inflate tree length or distort other metrics of fit. Line 264 refers to at least three independent origins of cephalic sutures in artiopodans but the fully resolved Figure 7c requires only two origins.*

We thank the reviewer for pointing this out. However now the analyses have been re-run we have new results to consider. The results still support multiple origins of sutures. We also note that the dots were indicating how terminals were coded. This is now clearer in the revised version of this figure (Figure 8 in the new version).

We have extended our interrogation of the trees by incorporating treespace analyses. These add support for the nodes of interest (around the base of trilobites), showing that the coding of *Acanthomeridion* ventral plate homologies impacts its position in the tree, and thus has implications for our understanding of the evolution of sutures in trilobites.

> *The question of how many times dorsal ecdysial sutures evolved in Artiopoda was addressed by Hou et al (2017), who first documented the facial sutures of Acanthomeridion and optimised them onto a phylogeny to infer multiple origins, as well as in a paper led by the lead author in Cladistics in 2019. Du et al. (2019) presented a phylogeny based on an earlier version of the current dataset wherein they discussed how many times sutures evolved or were lost based on their presence in*
>
> *Zhiwenia/Protosutura, Acanthomeridion, and Trilobita. To their credit, the authors acknowledge this (lines 62-65). The answer here is slightly different (because some topologies unite Acanthomeridion and trilobites).*
>
> *The following points are not meant to be "Weaknesses" but rather are refinements:*
>
> *I recommend changing the title of the paper from "cephalic sutures" to "dorsal ecdysial sutures" to be more precise about the character that is being tracked evolutionarily. Lots of arthropods have cephalic sutures (e.g., the ventral marginal suture of xiphosurans; the Y-shaped dorsomedian ecdysial line in insects). The text might also be updated to change other instances of "cephalic sutures" to a more precise wording.*

We appreciate this point and have changed the title as suggested.

> *The authors have provided (but not explicitly identified) support values for nodes in their Bayesian trees but not in their parsimony ones. Please do the jackknife or bootstrap for the parsimony analyses and make it clear that the Bayesian values are posterior probabilities.*

With the addition of further trilobite terminals to our parsimony analyses, the results became poor.

Specifically the internal relationships of trilobites did not conform to any previous study, and *Olenellus getzi* was not resolved as an early diverging member of the group. This meant that these analyses could not be used for addressing the hypothesis of reviewer two. We decided to exclude reporting parsimony analysis results from this version to avoid confusion.

We have added a note that the values reported at the nodes are posterior probabilities to figures S8, S9 and S10 where we show the full Bayesian results.

> *In line 65 or somewhere else, it might be noted that a single origin of the dorsal facial sutures in trilobites has itself been called into question. Jell (2003) proposed that separate lineages of Eutrilobita evolved their facial sutures independently from separate sister groups within Olenellina.*

We have added this to the introduction (Line 98). Thank you for raising this point.

> *I have provided minor typographic or terminological corrections to the authors in a list of recommendations that may not be publicly available.*

We appreciate the points made by the reviewer and their detailed corrections, which we have corrected in the revised version.

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